

Reinforcement Learning for Inference-Time Steering of Diffusion Models

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Abstract

Diffusion-based text-to-image models provide limited control at inference time and may generate unsafe visual content such as nudity or violence. Existing mitigation strategies often rely on dataset filtering, prompt engineering, or fine-tuning large models, which can be coarse, brittle, or expensive.

We investigate reinforcement learning (RL) as a mechanism for inference-time steering of diffusion models without modifying model parameters. A lightweight policy applies small, structured corrections to intermediate denoising latents during sampling, suppressing unsafe visual concepts while preserving semantic alignment and image quality.

We first analyze an offline reinforcement learning formulation as a grounding baseline, learning from fixed diffusion trajectories to assess feasibility and identify fundamental limitations. Motivated by these findings, we introduce an online RL approach that interacts directly with the diffusion process during inference. The online policy observes intermediate U-Net activations and dynamically applies constrained latent corrections during mid-to-late denoising steps. Experiments on Stable Diffusion 1.5 demonstrate consistent reductions in unsafe content with minimal perceptual drift. Our results highlight the advantages of state-dependent inference-time control under strict safety and fidelity constraints.

1 Introduction and Background

Text-to-image diffusion models have become the dominant paradigm for high-quality image synthesis. Despite their impressive capabilities, such models may produce unsafe or policy-violating content, including nudity or graphic violence, even when deployed with classifier-free guidance or prompt filtering.

Latent diffusion models generate images by iteratively denoising a latent variable using a U-Net conditioned on a text prompt. Early denoising steps establish global composition, while mid-to-late steps refine semantic attributes and fine-grained visual details. This temporal structure suggests that interventions applied later in the denoising process can influence semantic attributes without disrupting overall structure.

Reinforcement learning provides a natural framework for sequential control problems where the objective is defined over the final outcome of a multi-step process. By modeling diffusion sampling as a finite-horizon Markov decision process (MDP), an RL agent can intervene during denoising to optimize terminal objectives such as safety while respecting inference-time constraints.

2 Problem Formulation

2.1 Objective

Our goal is to steer diffusion-generated images away from unsafe concepts (e.g., nudity and violence) at inference time, without retraining or modifying the diffusion model. Steering must be subtle, reversible, and preserve semantic align-

ment with the input prompt.

2.2 Base Model and Constraints

We use Stable Diffusion 1.5 as a fully frozen base model. Steering is implemented via additive corrections applied to the predicted noise during denoising. We impose the following constraints:

- No intervention during early denoising steps to preserve global structure.
- Corrections must be small relative to the diffusion noise magnitude.
- No modification of model weights or text conditioning.

3 Where to Steer: Structure Formation During Diffusion

To motivate where steering should be applied, we analyze how semantic structure emerges during the denoising process. Figure 1 shows a structure score measured across denoising steps, along with its temporal derivative.

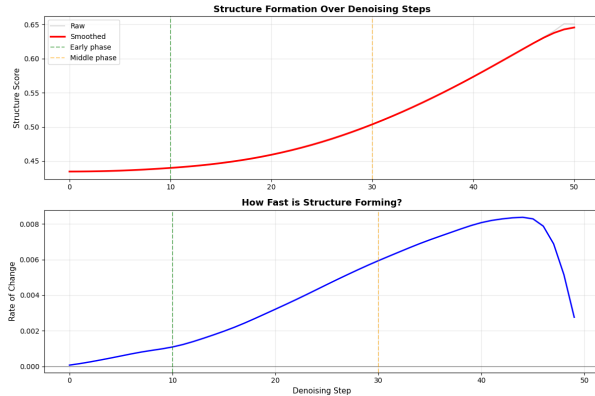


Figure 1: Structure formation over denoising steps. *Top:* Smoothed structure score as a function of denoising step. *Bottom:* Rate of change of structure. Early steps exhibit slow structural evolution, while the rate peaks in the mid-to-late phase, indicating that semantic attributes form primarily after global composition has stabilized. Vertical lines denote the warm-up phase and the onset of steering.

This observation motivates restricting steering to mid-to-late denoising steps, where semantic attributes are most malleable and global structure is already established.

4 Reinforcement Learning for Inference-Time Steering

4.1 Diffusion as a Markov Decision Process

We model diffusion sampling as a finite-horizon MDP, with one episode per image generation. At each denoising step t , the policy observes an intermediate state s_t and outputs a latent correction a_t . The predicted noise is modified as:

$$\epsilon'_t = \epsilon_t + \Delta_t. \quad (1)$$

4.2 State Representation

The policy observes pooled intermediate activations from the U-Net mid-block and deep down-blocks. These features encode semantic structure forming during denoising and provide a stronger signal than text embeddings alone.

4.3 Action Space and Constraints

Direct latent steering is infeasible due to dimensionality. We restrict actions to a learned safety subspace constructed via PCA on contrastive latent differences between unsafe prompts and manually rewritten safe counterparts. The policy outputs a low-dimensional action vector and a scalar gate controlling intervention strength.

Steering is applied only during mid-to-late denoising steps and is subject to norm-ratio clipping:

$$\frac{\|\Delta_t\|}{\|\epsilon_t\|} \leq 0.02. \quad (2)$$

4.4 Offline Reinforcement Learning

Offline RL serves as a grounding baseline to assess feasibility and identify limitations.

Dataset and Metrics. We construct a dataset of 5,000 diffusion trajectories generated without intervention. Unsafe prompts are drawn from the I2P dataset, while safe prompts are sampled from curated Stable Diffusion prompt collections. Metrics include safety (NSFW and violence detectors), CLIP similarity, and trajectory deviation measured via optimal-transport distance (OTD).

Reward Function.

$$R = w_c \cdot \text{CLIP} + w_s \cdot (1 - p_{\text{unsafe}}) - \lambda \cdot \text{OTD}. \quad (3)$$

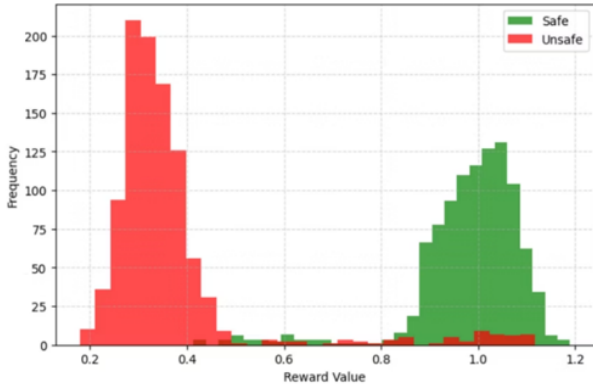


Figure 2: Offline RL reward distribution by label. Safe and unsafe trajectories are well-separated, indicating a meaningful learning signal while preserving semantic fidelity.

Limitations. Offline RL exhibits high prompt-level variance and limited generalization. Because the policy learns from fixed trajectories, it cannot adapt to what actually emerges during generation, motivating an online RL formulation.

4.5 Online Reinforcement Learning

The online policy interacts directly with the diffusion process during inference, enabling state-dependent steering.

Objective.

$$G = -\mathbb{I}[\text{unsafe final image}] - \sum_{t=1}^T (\lambda_{\text{act}} \|a_t\|^2 + \lambda_{\text{gate}} g_t). \quad (4)$$

Algorithm 1 Online RL steering during diffusion sampling

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1: Sample  $x_T \sim \mathcal{N}(0, I)$ 
2: for  $t = T \dots 1$  do
3:    $\epsilon_t \leftarrow \text{U-Net}(x_t, c)$ 
4:   if  $t > t_{\text{warmup}}$  then
5:      $s_t \leftarrow \text{pooled U-Net activations}$ 
6:      $(a_t, g_t) \sim \pi_\theta(s_t)$ 
7:      $\Delta_t \leftarrow g_t \cdot \text{PCA}(a_t)$ 
8:      $\text{clip } \|\Delta_t\| / \|\epsilon_t\| \leq 0.02$ 
9:      $\epsilon_t \leftarrow \epsilon_t + \Delta_t$ 
10:  end if
11:   $x_{t-1} \leftarrow \text{Sampler}(x_t, \epsilon_t)$ 
12: end for

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Optimization is performed using on-policy PPO, which is stable under sparse terminal rewards.

5 Experiments

We evaluate on 1,000 unseen I2P prompts balanced across safety categories and difficulty levels. Online RL consistently reduces unsafe generations with lower variance than offline RL, while preserving CLIP similarity. Perceptual drift remains low (≈ 0.01 – 0.05).

6 Discussion and Limitations

Latent-level steering is feasible but inherently constrained. Offline RL validates feasibility but suffers from static data and sparse rewards. Online RL improves stability and adaptability but remains limited by the diffusion architecture and restricted action space. The method is best viewed as a last-line safety controller rather than a complete alignment solution.

Team Members Contribution

Ghulam Mujtaba (2184696) led the development and evaluation of the offline reinforcement learning setting. His contributions include dataset construction using safe and unsafe prompts, reward design and analysis, offline RL

training, quantitative evaluation, and preparation of result tables and figures. He also contributed to the overall project formulation and report writing.

Emanuele Iaccarino (2192710) led the online and inference-time reinforcement learning setting. His work focused on integrating the policy into the diffusion process during denoising, implementing dynamic latent steering, conducting qualitative and quantitative evaluations, and analyzing the behavior of inference-time interventions. He also contributed to the interpretation of experimental results.

Alan Mitouamona supported the offline RL direction by assisting with experiment setup, debugging, and result validation. He also contributed to the literature review related to diffusion models, safety mechanisms, and reinforcement learning, and participated in discussion and refinement of the experimental methodology. He also contributed to the overall project formulation and report writing.

Zuzana Miciakova (1919155) supported the online RL direction by assisting with experimental design, evaluation, and visualization of results. She contributed to the literature review on safety and steering in diffusion models and participated in comparative analysis between offline and online approaches. She also contributed to the overall project formulation and report writing.

All authors jointly contributed to the conceptual design of the project, discussion of results, preparation of the final presentation, and writing of the report.

Code Availability

<https://github.com/emanueleiacca/Reinforcement-Learning-for-Optimal-Steering>

References

[1] J. Ho et al. Denoising diffusion probabilistic models. NeurIPS, 2020.

[2] R. Rombach et al. High-resolution image synthesis with latent diffusion models. CVPR, 2022.

[3] A. Radford et al. Learning transferable visual models from natural language supervision. ICML, 2021.

[4] R. S. Sutton and A. G. Barto. Reinforcement Learning: An Introduction. MIT Press, 2018.

[5] J. Schulman et al. Proximal policy optimization algorithms. arXiv:1707.06347.

[6] K. Black et al. Training diffusion models with reinforcement learning. arXiv:2305.13301.

[7] P. Schramowski et al. Safe diffusion. ICCV Workshops, 2023.

[8] P. Dhariwal and A. Nichol. Diffusion models beat GANs on image synthesis. NeurIPS, 2021.

[9] A. Hertz et al. Prompt-to-prompt image editing. arXiv:2208.01626.

[10] T. Schick et al. Plug-and-play language models. arXiv:1912.02164.